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TRANSPORT-BASED ENERGY ALLOCATION

Abstract

An example operation includes one or more of determining, via a transport, a state of charge of a storage unit of the transport, determining, via the transport, an environment at a location, determining, via the transport, a status of a charging station at the location including at least one of a location energy cost and a location power level and providing, via the transport, a needed amount of energy to the charging station at the location based on the status of the charging station.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/206,072, filed on Jun. 5, 2023, which is a continuation of U.S. patent application Ser. No. 16/821,974, filed on Mar. 17, 2020, now U.S. Pat. No. 11,685,283, issued on Jun. 27, 2023, the entire disclosures of which are incorporated herein by reference.

BACKGROUND

[0002] Currently, charging stations provide power from an electrical source provider to an electric transport. The charging station has little intelligence of the energy needs at that location nor of the energy available in transports in the vicinity of the charging station.

[0003] As such, what is sought is to provide charge to a location from a transport based on factors such as the temperature at the location, the cost of the energy and the power level at the location.

SUMMARY

[0004] One example embodiment provide a method that includes one or more of wirelessly notifying, via a charging station, at least one transport of a group of identified transports of a needed amount of energy stored in the at least one transport prior to the at least one transport being connected to the charging station.

[0005] Another example embodiment provides a system, comprising, a charging station configured to perform one or more of wirelessly notify at least one transport of a group of identified transports of a needed amount of energy stored in the at least one transport prior to the at least one transport being connected to the charging station.

[0006] A further example embodiment provides a non-transitory computer readable medium comprising instructions, that when read by a processor, causes the processor to perform one or more of wirelessly notifying, via a charging station, at least one transport of a group of identified transports of a needed amount of energy stored in the at least one transport prior to the at least one transport being connected to the charging station.

[0007] Another example embodiment provides a method that includes one or more of determining, via a transport, a state of charge of a storage unit of the transport, determining, via the transport, an environment at a location, determining, via the transport, a status of a charging station at the location including at least one of a location energy cost and a location power level and providing, via the transport, a needed amount of energy to the charging station at the location based on the status of the charging station.

[0008] A further example embodiment provides an apparatus, comprising, a charging station at a location that is configured to perform one or more of receive energy from a storage unit of a transport, and a processor configured to, determine, via the transport, a state of charge of the storage unit of the transport, determine, via the transport, an environment at a location, wirelessly determine, via the transport, a status of the charging station at the location including at least one of a location energy cost and a location power level and provide, via the transport, a needed amount of energy to the charging station at the location based on the status of the charging station.

[0009] Another example embodiment provides a non-transitory computer readable medium comprising instructions, that when read by a processor, cause the processor to perform one or more of determining, via a transport, a state of charge of a storage unit of the transport, determining, via the transport, an environment at a location, determining, via the transport, a status of a charging station at the location including at least one of a location energy cost and a location power level and

providing, via the transport, a needed amount of energy to the charging station at the location based on the status of the charging station.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- [0010] FIG. 1A illustrates a first example power return notification system overview, according to example embodiments.
- [0011] FIG. 1B illustrates a second example power return notification system overview, according to example embodiments.
- [0012] FIG. 1C illustrates an example vehicle-to-vehicle power transfer notification before the transfer, according to example embodiments.
- [0013] FIG. 1D illustrates an example vehicle-to-vehicle power transfer notification during the transfer, according to example embodiments.
- [0014] FIG. 1E illustrates an example vehicle-to-location power transfer, according to example embodiments.
- [0015] FIG. 2A illustrates a transport network diagram, according to example embodiments.
- [0016] FIG. 2B illustrates another transport network diagram, according to example embodiments.
- [0017] FIG. 2C illustrates yet another transport network diagram, according to example embodiments.
- [0018] FIG. 2D illustrates another transport network diagram, according to example embodiments.
- [0019] FIG. 3A illustrates a first flow diagram, according to example embodiments.
- [0020] FIG. 3B illustrates a second flow diagram, according to example embodiments.
- [0021] FIG. 4 illustrates a machine learning transport network diagram, according to example embodiments.
- [0022] FIG. 5A illustrates an example vehicle configuration for managing database transactions associated with a vehicle, according to example embodiments.
- [0023] FIG. 5B illustrates another example vehicle configuration for managing database transactions conducted among various vehicles, according to example embodiments.
- [0024] FIG. 6A illustrates a blockchain architecture configuration, according to example embodiments.
- [0025] FIG. 6B illustrates another blockchain configuration, according to example embodiments.
- [0026] FIG. 6C illustrates a blockchain configuration for storing blockchain transaction data, according to example embodiments.
- [0027] FIG. 6D illustrates example data blocks, according to example embodiments.
- [0028] FIG. 7 illustrates an example system that supports one or more of the example embodiments.

DETAILED DESCRIPTION

[0029] It will be readily understood that the instant components, as generally described and illustrated in the figures herein, may be arranged and designed in a wide variety of different configurations. Thus, the following detailed description of the embodiments of at least one of a method, apparatus, non-transitory computer readable medium and system, as represented in the attached figures, is not intended to limit the scope of the application as claimed but is merely representative of selected embodiments.

[0030] The instant features, structures, or characteristics as described throughout this specification may be combined in any suitable manner in one or more embodiments. For example, the usage of the phrases “example embodiments”, “some embodiments”, or other similar language, throughout least this specification refers to the fact that a particular feature, structure, or characteristic described in connection with the embodiment may be included in at one embodiment. Thus,

appearances of the phrases “example embodiments”, “in some embodiments”, “in other embodiments”, or other similar language, throughout this specification do not necessarily all refer to the same group of embodiments, and the described features, structures, or characteristics may be combined in any suitable manner in one or more embodiments. In the diagrams, any connection between elements can permit one-way and/or two-way communication even if the depicted connection is a one-way or two-way arrow. In the current solution, a transport may include one or more of vehicles, cars, trucks, motorcycles, scooters, bicycles, boats, recreational vehicles, planes, and any object that may be used to transport people and or goods from one location to another.

[0031] In addition, while the term “message” may have been used in the description of embodiments, the application may be applied to many types of network data, such as, a packet, frame, datagram, etc. The term “message” also includes packet, frame, datagram, and any equivalents thereof. Furthermore, while certain types of messages and signaling may be depicted in exemplary embodiments they are not limited to a certain type of message, and the application is not limited to a certain type of signaling.

[0032] Example embodiments provide methods, systems, components, non-transitory computer readable media, devices, and/or networks, which provide at least one of: a transport (also referred to as a vehicle herein) a data collection system, a data monitoring system, a verification system, an authorization system and a vehicle data distribution system. The vehicle status condition data, received in the form of communication update messages, such as wireless data network communications and/or wired communication messages, may be received and processed to identify vehicle/transport status conditions and provide feedback as to the condition changes of a transport. In one example, a user profile may be applied to a particular transport/vehicle to authorize a current vehicle event, service stops at service stations, and to authorize subsequent vehicle rental services.

[0033] Within the communication infrastructure, a decentralized database is a distributed storage system which includes multiple nodes that communicate with each other. A blockchain is an example of a decentralized database which includes an append-only immutable data structure (i.e. a distributed ledger) capable of maintaining records between untrusted parties. The untrusted parties are referred to herein as peers, nodes or peer nodes. Each peer maintains a copy of the database records and no single peer can modify the database records without a consensus being reached among the distributed peers. For example, the peers may execute a consensus protocol to validate blockchain storage entries, group the storage entries into blocks, and build a hash chain via the blocks. This process forms the ledger by ordering the storage entries, as is necessary, for consistency. In a public or permissionless blockchain, anyone can participate without a specific identity. Public blockchains can involve cryptocurrencies and use consensus based on various protocols such as proof of work (PoW). On the other hand, a permissioned blockchain database provides a system which can secure interactions among a group of entities which share a common goal, but which do not or cannot fully trust one another, such as businesses that exchange funds, goods, information, and the like. The instant application can function in a permissioned and/or a permissionless blockchain setting.

[0034] Smart contracts are trusted distributed applications which leverage tamper-proof properties of the shared or distributed ledger (i.e., which may be in the form of a blockchain) database and an underlying agreement between member nodes which is referred to as an endorsement or endorsement policy. In general, blockchain entries are “endorsed” before being committed to the blockchain while entries which are not endorsed are disregarded. A typical endorsement policy allows smart contract executable code to specify endorsers for an entry in the form of a set of peer nodes that are necessary for endorsement. When a client sends the entry to the peers specified in the endorsement policy, the entry is executed to validate the entry. After validation, the entries enter an ordering phase in which a consensus protocol is used to produce an ordered sequence of endorsed entries grouped into blocks.

[0035] Nodes are the communication entities of the blockchain system. A “node” may perform a

logical function in the sense that multiple nodes of different types can run on the same physical server. Nodes are grouped in trust domains and are associated with logical entities that control them in various ways. Nodes may include different types, such as a client or submitting-client node which submits an entry-invocation to an endorser (e.g., peer), and broadcasts entry-proposals to an ordering service (e.g., ordering node). Another type of node is a peer node which can receive client submitted entries, commit the entries and maintain a state and a copy of the ledger of blockchain entries. Peers can also have the role of an endorser, although it is not a requirement. An ordering-service-node or orderer is a node running the communication service for all nodes, and which implements a delivery guarantee, such as a broadcast to each of the peer nodes in the system when committing entries and modifying a world state of the blockchain, which is another name for the initial blockchain entry which normally includes control and setup information.

[0036] A ledger is a sequenced, tamper-resistant record of all state transitions of a blockchain. State transitions may result from smart contract executable code invocations (i.e., entries) submitted by participating parties (e.g., client nodes, ordering nodes, endorser nodes, peer nodes, etc.). An entry may result in a set of asset key-value pairs being committed to the ledger as one or more operands, such as creates, updates, deletes, and the like. The ledger includes a blockchain (also referred to as a chain) which is used to store an immutable, sequenced record in blocks. The ledger also includes a state database, which maintains a current state of the blockchain. There is typically one ledger per channel. Each peer node maintains a copy of the ledger for each channel of which they are a member.

[0037] A chain is an entry log which is structured as hash-linked blocks, and each block contains a sequence of N entries where N is equal to or greater than one. The block header includes a hash of the block's entries, as well as a hash of the prior block's header. In this way, all entries on the ledger may be sequenced and cryptographically linked together. Accordingly, it is not possible to tamper with the ledger data without breaking the hash links. A hash of a most recently added blockchain block represents every entry on the chain that has come before it, making it possible to ensure that all peer nodes are in a consistent and trusted state. The chain may be stored on a peer node file system (i.e., local, attached storage, cloud, etc.), efficiently supporting the append-only nature of the blockchain workload.

[0038] The current state of the immutable ledger represents the latest values for all keys that are included in the chain entry log. Because the current state represents the latest key values known to a channel, it is sometimes referred to as a world state. Smart contract executable code invocations execute entries against the current state data of the ledger. To make these smart contract executable code interactions efficient, the latest values of the keys may be stored in a state database. The state database may be simply an indexed view into the chain's entry log, it can therefore be regenerated from the chain at any time. The state database may automatically be recovered (or generated if needed) upon peer node startup, and before entries are accepted.

[0039] A blockchain is different from a traditional database in that the blockchain is not a central storage but rather a decentralized, immutable, and secure storage, where nodes must share in changes to records in the storage. Some properties that are inherent in blockchain and which help implement the blockchain include, but are not limited to, an immutable ledger, smart contracts, security, privacy, decentralization, consensus, endorsement, accessibility, and the like.

[0040] Example embodiments provide a way for providing a vehicle service to a particular vehicle and/or requesting user associated with a user profile that is applied to the vehicle. For example, a user may be the owner of a vehicle or the operator of a vehicle owned by another party. The vehicle may require service at certain intervals and the service needs may require authorization prior to permitting the services to be received. Also, service centers may offer services to vehicles in a nearby area based on the vehicle's current route plan and a relative level of service requirements (e.g., immediate, severe, intermediate, minor, etc.). The vehicle needs may be monitored via one or more sensors which report sensed data to a central controller computer device in the vehicle, which

in turn, is forwarded to a management server for review and action.

[0041] A sensor may be located on one or more of the interior of the transport, the exterior of the transport, on a fixed object apart from the transport, and on another transport near to the transport. The sensor may also be associated with the transport's speed, the transport's braking, the transport's acceleration, fuel levels, service needs, the gear-shifting of the transport, the transport's steering, and the like. The notion of a sensor may also be a device, such as a mobile device. Also, sensor information may be used to identify whether the vehicle is operating safely and whether the occupant user has engaged in any unexpected vehicle conditions, such as during the vehicle access period. Vehicle information collected before, during and/or after a vehicle's operation may be identified and stored in a transaction on a shared/distributed ledger, which may be generated and committed to the immutable ledger as determined by a permission granting consortium, and thus in a "decentralized" manner, such as via a blockchain membership group. Each interested party (i.e., company, agency, etc.) may want to limit the exposure of private information, and therefore the blockchain and its immutability can limit the exposure and manage permissions for each particular user vehicle profile. A smart contract may be used to provide compensation, quantify a user profile score/rating/review, apply vehicle event permissions, determine when service is needed, identify a collision and/or degradation event, identify a safety concern event, identify parties to the event and provide distribution to registered entities seeking access to such vehicle event data. Also, the results may be identified, and the necessary information can be shared among the registered companies and/or individuals based on a "consensus" approach associated with the blockchain. Such an approach could not be implemented on a traditional centralized database.

[0042] In an embodiment of the current solution, the charging station takes on an additional role and/or a different role as a wireless communicator with transports. The disclosed charging stations communicate the power needs of the system(s) in which it is in communication with (or the needs of the charging station(s) themselves), receives energy from the transport for delivery to the local electrical grid or locale and is aware of the locations, direction of travel and status of various electric transports.

[0043] Another embodiment of the current solution wirelessly notifies a transport to provide a portion of an amount of energy prior to the coupling of the transport and charging station, and provides intelligence between the charging station and the transport, allowing the processor of the charging station to wireless communicate with the transport regarding an amount of energy store in a battery/batteries on the transport. The processor and/or other devices such as a transceiver, transmitter, receiver, sensor and the like of the charging station is configured to communicate with one or more processors and/or other devices such as a transceiver, transmitter, receiver, sensor and the like of the transport to determine a current battery charge, and to provide a fraction of the charge to the charging station. In one example the coupling of the transport and the charging station may be by way of direct electrical connection, inductive coupling and the like.

[0044] In an embodiment of the current solution, the charging station acts as a vehicle energy director and a bi-directional energy transit/transfer device in which energy needs are met both from the electrical grid to an endpoint and from an endpoint to the electrical grid. The endpoint may be a vehicle, a dwelling, another charging station, and the like. This bi-directionality of energy flow is performed by the current solution that has expanded intelligence into the transports surrounding it. The expanded intelligence of the charging station may include transport status information such as the current state of charge of the transport, its location, direction of travel, endpoint, other stops, its current usage rate, and the like received via wireless communication from the transport. The charging station may determine an additional amount of energy that the transport will require to complete its route, this information may also be wirelessly communicated by the transport to the charging station as transport status information. The charging station may additionally store previous transport status information from previous interactions with the transport and based on these previous interactions form a historical travel pattern for that transport.

[0045] The charging stations may communicate with one another to provide information related to a direction of transit travel, state of charge, a transit time to the charging station(s), etc. In this current solution, the charging stations, in one embodiment, can communicate with one another as well as transiting vehicles, keeping track of their energy needs and excess energy storage capacity. This information is used to select specific vehicle(s) based on a current energy storage of the vehicle(s) and a time to a charging station. In other embodiments, an estimated time that will be spent by the vehicle discharging at the charging station may also be determined and used to select the specific vehicle(s) as well as to manage the ingress and egress of vehicle(s) at the charging station to provide additional power to the electrical grid (FIG. 1A, 142). The estimated time spent by a vehicle discharging at the charging station may be based on the state of charge of the transport, the type of transport, the type of energy storage, the type of electrical current connection and the like. The communication between charging stations may be one of wireless, wired, network, internet based and the like.

[0046] In one example embodiment, the charging station wirelessly notifies a transport of a group of transports about which it is aware, traveling toward it and having excess energy to provide a needed amount of energy that the transport has and can spare to the charging station.

[0047] In one embodiment the charging station may wirelessly communicate with and may be aware of the group of electric transports in its vicinity, their direction of travel and their state of charge of the individual transiting transport batteries. In one embodiment the transport storage unit may be comprised of capacitors, super-capacitors and the like. In one embodiment, the transports that are in wireless communication with the charging station send status information to the charging station pertaining to the vehicle including one or more of: its current state, location, travel itinerary, load, energy usage per mile and the like. The charging station gathers the status information to form a status matrix of the transports that the charging station is in communication with. This status matrix may be used as the basis of decisions made by the charging station about the group of transports traveling toward the charging station. The status matrix may be utilized to select potential candidates to transfer some of their energy to the charging station. The status matrix may be stored at the charging station, in the cloud or on a server communicably coupled to the charging station. The status information associated with the transport may be stored locally in the transport, in the cloud or on a server communicably coupled to the transport.

[0048] In one embodiment a status matrix is maintained by the charging station. In this embodiment, the status matrix is a matrix of information for transports approaching the charging station. In other embodiments, the status matrix is maintained while the transport is within wireless communication range of the charging station or while the transport is within a distance of the charging station. This matrix of information of the transports in communication with the charging station may be utilized for decisions pertaining to energy transfer and a history of the matrix of information for the transports may be kept for subsequent historical analysis. The information maintained by the status matrix may include one or more of a state of charge, a travel direction, endpoint(s), a current location, road blockages at the location, a travel itinerary, a load, an initiation point, an endpoint, an instantaneous speed of travel, an average speed of travel, a top speed, an acceleration rate, a current energy usage rate per distance (such as mile or kilometer), an amount of charge to provide, an amount of time to spend at the charging station, current and/or future road conditions, current and/or future weather conditions, an estimated distance to the charging station, an estimated distance to a subsequent endpoints, etc. This information can be utilized to make decisions by the charging station (via one or more processors, sensors and/or memories, which can store the status matrix and/or information on the charging station or accessible by the charging station) for one or more of the transports in wireless communication with the charging station. The status matrix may also include historical travel patterns and may be based on previously stored transport status reports from previous interactions, the probability of the transport traveling directly to an endpoint on a particular day and/or stops along the way, etc.

[0049] In one example embodiment in which the status matrix has a minimal number of components, the matrix may include the identifiers for each transport it is communication with, their location with respect to the charging station, whether the transport is traveling toward the charging station, the state of charge of the transport and one or more endpoints of the transport. With that information in the matrix, a determination may be made as to which vehicle traveling toward the charging station will have the most excess energy after factoring in the energy needed to travel from the charging station to the endpoint. The transport with the greatest excess energy may be selected to transfer a portion of its excess energy back to the charging station and from there, optionally, to the electrical grid or locale. Other data may be collected as discussed previously and a different manner of selection of a transport for providing energy may be utilized.

[0050] The charging station may select potential candidates based on those transports with the largest energy reserves, the largest ratio of energy reserves to estimated energy usage, transports closest to their endpoint, transports approaching the charging station and near their endpoint, transports that will have the shortest queue time based on a current speed and energy download/expelling time of a transport and the like. The endpoint may be a vehicle, a dwelling, another charging station, and the like.

[0051] The charging station will not leave the transport energy deficient and unable to complete its journey because the charging station will have either the currently planned route of the transport via wireless transmission from the transport or access to its historical travel patterns via data stored in the transport, data stored about the transport in the charging station, data stored about the transport in the cloud, a server and the like. For example, the historical travel pattern may be based on previously stored transport status reports from previous interactions, the charging station may determine the probability of the transport traveling directly to a location on a particular day, or stops along the way, etc. This information may form a portion of the status matrix that may be utilized by the charging station to determine which vehicle(s) will form the group.

[0052] The transport wirelessly communicates with the charging station and provides one or more of its state of charge, travel direction, endpoint(s), location, an amount of charge to provide, an amount of time to spend at the charging station, and estimated distance to the charging station, estimate distance to a subsequent endpoint, etc. The transport may independently confirm how much energy it can spare.

[0053] In one embodiment, the charging station wirelessly communicates its energy need to the transport and a backup transport in case the first transport is not able or willing to transfer energy to the charging station. The system may also schedule the communications and energy transfer so that the transferring transport is at the charging station for the least amount of time in queue. High efficiency of transport energy upload throughput with minimal queue times for both the charging station and transports may be achieved by scheduling the transport for energy transfer based on a distance from the transport to the charging station and the instantaneous velocities of transports traveling toward the charging station based on their wirelessly communicated status information. The queue times may also be reduced by determining the travel time of prior transports at approximately the same distance from the charging station and scheduling the transport to arrive as the previous transport is completing its energy transfer. In some embodiments the travel times may be a function of the time of day, distance, traffic density, traffic throughput hindrances and the like.

[0054] FIG. 1A illustrates an example power return notification system overview **100**. In one example embodiment a charging station (FIG. 1A, **114**) is configured to wirelessly notify at least one transport (FIG. 1A, **110**) of a group of identified transports (FIG. 1A, **110** and **112**) of a needed amount of energy stored in the at least one transport (FIG. 1A, **110**) prior to the at least one transport being connected to the charging station (FIG. 1A, **114**).

[0055] FIG. 1B illustrates another example power return notification system overview **120**. In another example, a charging station (FIG. 1B, **114**) is configured to receive energy from a transport storage unit (FIG. 1B, **134**, **136**). A processor wirelessly may notify at least one transport (FIG. 1B,

110) of a group of identified transports (FIG. **1B**, **110** and **112**) of an indication of a needed amount of energy stored in the at least one transport (FIG. **1B**, **110**) prior to the at least one transport (FIG. **1B**, **110**) being connected to the charging station (FIG. **1B**, **114**).

[0056] In yet another example embodiment, the charging station (FIG. **1B**, **114**) may wirelessly notify another transport (FIG. **1B**, **112**) of the group of identified transports (FIG. **1B**, **110** and **112**) of the needed amount of energy stored in the other transport (FIG. **1B**, **112**). The charging station (FIG. **1B**, **114**) may also direct the another transport (FIG. **1B**, **112**) to provide a portion of the amount of the energy to the charging station (FIG. **1B**, **114**) prior to the at least one transport (FIG. **1B**, **112**) being connected to the charging station (FIG. **1B**, **114**).

[0057] FIG. **1C** illustrates an example vehicle-to-vehicle power transfer notification before the transfer **150**. An example embodiment may include a charging station (FIG. **1C**, **114**) and a processor that determines an amount of stored energy in at least one storage unit (FIG. **1B**, **134**) of a transport (FIG. **1C**, **110**). The processor may determine an amount of reserved energy to perform a set of transport tasks, determine an amount of available energy based on the amount of stored energy and a reserve energy and a discharge limit of the at least one storage unit (FIG. **1B**, **134**). The processor may provide an indication of the amount of available energy of the at least one storage unit (FIG. **1B**, **134**) of a transport and based on the indication, provide notification that a portion of the available energy is to be provided to the charging station (FIG. **1B**, **134**) prior to being coupled.

[0058] In other embodiments, the system may provide a location of the charging station to the at least one transport when at least one of a following occurs, the charging station wirelessly notifies the at least one transport of the needed amount of energy and the at least one transport confirms the needed amount of energy can be provided to the charging station. The charging station may wirelessly notify another transport of the group of identified transports of the needed amount of energy stored in the other transport prior to the at least one transport being connected to the charging station. The charging station may wirelessly notify the transport of the needed amount of energy stored in the transport based on an amount of energy stored in the transport and a time for the transport to arrive at the charging station.

[0059] The charging station may query, via the charging station, at least one transport of an amount of energy stored in the at least one transport and provide, via the at least one transport to the charging station, the amount of energy stored in the at least one transport. The charging station may also identify the group of identified transports based on a distance, speed, direction, an amount of energy stored in each of the group of identified transports, an estimated amount of energy stored in each of the group of identified transports when each of the group of identified transports arrives at the charging station, and an estimated amount of energy stored in each of the group of identified transports when each of the group of identified transports arrives at the charging station and can provide energy to the charging station.

[0060] The charging station (FIG. **1C**, **114**) may possess information about the energy status of each transport in its immediate vicinity and may inform one energy-rich transport (FIG. **1C**, **110**) of another transport near the energy-rich transport that is energy-depleted (FIG. **1C**, **112**) and does not have sufficient energy to reach its endpoint.

[0061] FIG. **1D** illustrates another example vehicle-to-vehicle power transfer notification before the transfer **160**. In this embodiment, the charging system may schedule, but not directly participate in, a roadside transfer of energy from the energy-rich transport (FIG. **1D**, **110**) to the energy-depleted transport (FIG. **1D**, **112**), wherein the energy-depleted transport is not part of the group of identified transports. The roadside transfer may occur via charging cables connecting the energy-rich transport to the energy-depleted transport. In this case, neither the energy-rich nor the energy-depleted transport would be connected to the charging station, but would be directed by the charging station to perform an action so that both transports would be able to complete their journeys with no impact to the electrical grid (FIG. **1A**, **142**). In one example embodiment the

coupling of the energy-rich transport and the energy-poor transport may be by way of direct electrical connection, inductive coupling and the like.

[0062] FIG. 1E illustrates a further example vehicle-to-vehicle power transfer notification before the transfer 175. In one example an apparatus, comprising, a charging station (FIG. 1C, 114) at a location (FIG. 1E, 176) that is configured to receive energy from a storage unit (FIG. 1B, 134) of a transport (FIG. 1B, 114). The apparatus has a processor configured to determine, via the transport, a state of charge of the storage unit of the transport. The transport (FIG. 1B, 110), communicates wirelessly (FIG. 1B, 116) with the charge station (FIG. 1B, 114) and communicates the state of charge of the storage unit (FIG. 1B, 134) of the transport and the location of the transport via a spatial location sensor (FIG. 1B, 128, 132) via a transceiver (FIG. 1B, 124, 126). The processor also determines, via the transport, an environment at a location. In one example the temperature inside the location is measured and fed back to the transport or may be determined by an external temperature in the area. The processor may wirelessly determine, via the transport, a status of the charging station at the location including at least one of a location energy cost and a location power level. The charging station may feedback to the transport via communications (FIG. 1B, 116) a power level of the charging station (FIG. 1B, 116). The transport (FIG. 1B, 110) may receive the communications via the transceiver (FIG. 1B, 124) at the transport. The charging station status may be one of a power grid status or an internal power status of the charging station, and the like. The processor may also provide, via the transport, a needed amount of energy to the charging station at the location based on the status of the charging station. In one embodiment the charging station aware of its power need and the availability of power of the transport may receive from the transport a needed amount of energy.

[0063] In other embodiments, the environment at the location may be based on of an external temperature of the transport and a geographic position of the location. The apparatus may include a temperature sensor at the transport and be aware of the geolocation of the charging station at the location. Based on the distance and specific geolocation parameters, the environment may be determined. The environment at the location may be based on a current seasonal shading of the location and a current weather condition. The apparatus may also be aware of the date, and based in part on this may assign a current seasonal shading of the location, the position of the location in relation to other buildings and trees may also have an effect on the shading of the location. In addition to a temperature at the location, the weather such as raining and or wind may have an effect on the temperature of the location. The location may be any type of residence, dwelling or place where humans may be located such as a house, condo, townhome, hotel, office, business, warehouse, and the like. The environment may be within the location, a room within the location, an area within the location, around the area or external to the location. The environment of the location may also be affected by the number of people at the location. The temperatures may be either estimated or actual at both the transport and the location.

[0064] The transport may determine the estimated temperature of the location by determining the temperature of the vehicle at that moment and imputing a temperature delta based on historical precedent, or, the transport may look at the current weather conditions for the specific zip code identifier and upload the data from the internet. The transport may also estimate the temperature at the location by requesting the temperature of other vehicles in the vicinity of the location at that point in time, or, the transport may access weather stations in the vicinity of the location, or, the transport may access historical records for that date and the temperature at that date and offset the temperature bases on the delta from the historical temperature capture to the current temperature, and the like.

[0065] The processor may further optimally schedule, via the transport, receipt of the needed amount of energy to the charging station from the transport based on the location energy cost. An optimal scheduling of the additional energy may be related to scheduling the additional energy based on an estimate of when the energy will be needed and the cost of the energy at that moment

in time. Scheduling the transport to arrive just as electrical rates increase and the energy need increases may be optimally scheduled. The processor may determine, via the transport, the needed amount of energy to the charging station at the location based on one or more of an external temperature at the transport, the estimated temperature in the location, and the location energy cost. In an alternative embodiment, the temperature of the location may be an actual temperature conveyed via an internet of things temperature sensor, a local temperature grid, and the like. The temperature at the location may be estimated based on a position of the location within the local temperature grid. The transport may determine a state of charge of the charging station at the location. The state of charge of the charging station may be based on local power storage voltages and currents from storage devices such as batteries, capacitors and super-capacitors at the charging station. The processor may also execute, via the transport, a smart contract between the transport and the charging station related to the needed amount of energy.

[0066] The transport may determine the state of charge of the charging station at the location by querying the charging station directly, or, the transport may query a vehicle currently at the charging station, or, the transport may query a vehicle that has just interacted with the charging station. Additionally, the transport may determine the state of charge of the charging station by querying another charging station as to the state of charge of the charging station at the location, and the like.

[0067] FIG. 2A illustrates a transport network diagram **200**, according to example embodiments. The network comprises elements including a transport node **202** including a processor **204**, as well as a transport node **202'** including a processor **204'**. The transport nodes **202**, **202'** communicate with one another via the processors **204**, **204'**, as well as other elements (not shown) including transceivers, transmitters, receivers, storage, sensors and other elements capable of providing communication. The communication between the transport nodes **202**, **202'** can occur directly, via a private and/or a public network (not shown) or via other transport nodes and elements comprising one or more of a processor, memory, and software. Although depicted as single transport nodes and processors, a plurality of transport nodes and processors may be present. One or more of the applications, features, steps, solutions, etc., described and/or depicted herein may be utilized and/or provided by the instant elements.

[0068] FIG. 2B illustrates another transport network diagram **210**, according to example embodiments. The network comprises elements including a transport node **202** including a processor **204**, as well as a transport node **202'** including a processor **204'**. The transport nodes **202**, **202'** communicate with one another via the processors **204**, **204'**, as well as other elements (not shown) including transceivers, transmitters, receivers, storage, sensors and other elements capable of providing communication. The communication between the transport nodes **202**, **202'** can occur directly, via a private and/or a public network (not shown) or via other transport nodes and elements comprising one or more of a processor, memory, and software. The processors **204**, **204'** can further communicate with one or more elements **230** including sensor **212**, wired device **214**, wireless device **216**, database **218**, mobile phone **220**, transport node **222**, computer **224**, I/O device **226** and voice application **228**. The processors **204**, **204'** can further communicate with elements comprising one or more of a processor, memory, and software.

[0069] Although depicted as single transport nodes, processors and elements, a plurality of transport nodes, processors and elements may be present. Information or communication can occur to and/or from any of the processors **204**, **204'** and elements **230**. For example, the mobile phone **220** may provide information to the processor **204** which may initiate the transport node **202** to take an action, may further provide the information or additional information to the processor **204'** which may initiate the transport node **202'** to take an action, may further provide the information or additional information to the mobile phone **220**, the transport node **222**, and/or the computer **224**. One or more of the applications, features, steps, solutions, etc., described and/or depicted herein may be utilized and/or provided by the instant elements.

[0070] FIG. 2C illustrates yet another transport network diagram **240**, according to example embodiments. The network comprises elements including a transport node **202** including a processor **204** and a non-transitory computer readable medium **242C**. The processor **204** is communicably coupled to the non-transitory computer readable medium **242C** and elements **230** (which were depicted in FIG. 2B).

[0071] The processor **204** wirelessly notifies **244C**, via a charging station, at least one transport of a group of identified transports of a needed amount of energy stored in the at least one transport prior to the at least one transport being connected to the charging station.

[0072] In other embodiments, the processor may provide a location of the charging station to the at least one transport when at least one of a following occurs, the charging station wirelessly notifies the at least one transport of the needed amount of energy and the at least one transport confirms the needed amount of energy can be provided to the charging station. The processor may wirelessly notify, via the charging station, at least one other transport of the group of identified transports of the needed amount of energy stored in the at least one other transport prior to the at least one transport being connected to the charging station. The processor may also wirelessly notify, via the charging station, the at least one transport of the needed amount of energy stored in the at least one transport based on an amount of energy stored in the at least one transport and a time for the at least one transport to arrive at the charging station. The processor may also query, via the charging station, the at least one transport of an amount of energy stored in the at least one transport and provide, via the at least one transport to the charging station, the amount of energy stored in the at least one transport. The processor may also identify, via the charging station, the group of identified transports based on a distance, speed, direction, an amount of energy stored in each of the group of identified transports, an estimated amount of energy stored in each of the group of identified transports when each of the group of identified transports arrives at the charging station, and an estimated amount of energy stored in each of the group of identified transports when each of the group of identified transports arrives at the charging station and can provide energy to the charging station.

[0073] In another example, a non-transitory computer readable medium comprising instructions, that when read by a processor, causes the processor to perform, providing a wireless notification to at least one transport of a group of identified transports of an indication of a needed amount of energy stored in the at least one transport prior to the at least one transport being connected to a charging station and providing a wireless notification to at least one other transport of the group of identified transports of an indication of a needed amount of energy stored in the at least one other transport to provide a portion of the amount of the energy to the charging station prior to the at least one transport being connected to the charging station.

[0074] FIG. 2D illustrates yet another transport network diagram **240**, according to example embodiments. The network comprises elements including a transport node **202** including a processor **204** and a non-transitory computer readable medium **262C**. The processor **204** is communicably coupled to the non-transitory computer readable medium **262C** and elements **230** (which were depicted in FIG. 2B).

[0075] The processor **204** determines **264C**, via a transport, a state of charge of a storage unit of the transport, determines **266C**, via the transport, an environment at a location, determines **268C**, via the transport, a status of a charging station at the location including at least one of a location energy cost and a location power level and provides **270C**, via the transport, a needed amount of energy to the charging station at the location based on the status of the charging station.

[0076] In other embodiments, the environment at the location may be based on of an external temperature of the transport and a geographic position of the location. The environment at the location may be based on a current seasonal shading of the location and a current weather condition.

[0077] In other embodiments, the processor may optimally schedule, via the transport, receipt of

the needed amount of energy to the charging station from the transport based on the location energy cost, and/or determine, via the transport, the needed amount of energy to the charging station at the location based on one or more of an external temperature at the transport, the estimated temperature in the location and the location energy cost. The processor may also determine, via the transport, a state of charge of the charging station at the location and/or execute, via the transport, a smart contract between the transport and the charging station related to the needed amount of energy.

[0078] The processors and/or computer readable media may fully or partially reside in the interior or exterior of the transport nodes. The steps or features stored in the computer readable media may be fully or partially performed by any of the processors and/or elements in any order. Additionally, one or more steps or features may be added, omitted, combined, performed at a later time, etc.

[0079] FIG. 3A illustrates a flow diagram **300**, according to example embodiments. Referring to FIG. 3A, the flow comprises wirelessly notifying **302**, via a charging station, at least one transport of a group of identified transports of a needed amount of energy stored in the at least one transport prior to the at least one transport being connected to the charging station.

[0080] In other embodiments, the method may provide a location of the charging station to the at least one transport when at least one of a following occurs, the charging station wirelessly notifies the at least one transport of the needed amount of energy and the at least one transport confirms the needed amount of energy can be provided to the charging station. The method may also wirelessly notify, via the charging station, at least one other transport of the group of identified transports of the needed amount of energy stored in the at least one other transport prior to the at least one transport being connected to the charging station and may wirelessly notify, via the charging station, the at least one transport of the needed amount of energy stored in the at least one transport based on an amount of energy stored in the at least one transport and a time for the at least one transport to arrive at the charging station. The processor may also query, via the charging station, the at least one transport of an amount of energy stored in the at least one transport and provide, via the at least one transport to the charging station, the amount of energy stored in the at least one transport. The processor may also identify, via the charging station, the group of identified transports based on a distance, speed, direction, an amount of energy stored in each of the group of identified transports, an estimated amount of energy stored in each of the group of identified transports when each of the group of identified transports arrives at the charging station, and an estimated amount of energy stored in each of the group of identified transports when each of the group of identified transports arrives at the charging station and can provide energy to the charging station.

[0081] In another example, a method includes wirelessly notifying, via a processor, at least one transport of a group of identified transports of an indication of a needed amount of energy stored in the at least one transport prior to the at least one transport being connected to a charging station. The method may also include wirelessly notifying, via the processor, at least one other transport of the group of identified transports of an indication of a needed amount of energy stored in the at least one other transport to provide a portion of the amount of the energy to the charging station prior to the at least one transport being connected to the charging station.

[0082] FIG. 3B illustrates a flow diagram **310**, according to example embodiments. Referring to FIG. 3B, the flow comprises determining **312**, via a transport, a state of charge of a storage unit of the transport, determining **314**, via the transport, an environment at a location, determining **316**, via the transport, a status of a charging station at the location including at least one of a location energy cost and a location power level and providing **318**, via the transport, a needed amount of energy to the charging station at the location based on the status of the charging station.

[0083] In other embodiments, the environment at the location may be based on of an external temperature of the transport and a geographic position of the location. The environment at the location may be based on a current seasonal shading of the location and a current weather condition.

[0084] The method may further include optimally scheduling, via the transport, receipt of the needed amount of energy to the charging station from the transport based on the location energy cost, and/or determining, via the transport, the needed amount of energy to the charging station at the location based on one or more of an external temperature at the transport, the estimated temperature in the location, and the location energy cost. The method may also include determining, via the transport, a state of charge of the charging station at the location and/or executing, via the transport, a smart contract between the transport and the charging station related to the needed amount of energy.

[0085] FIG. 4 illustrates a machine learning transport network diagram **400**, according to example embodiments. The network **400** includes a transport node **402** that interfaces with a machine learning subsystem **406**. The transport node includes one or more sensors **404**.

[0086] The machine learning subsystem **406** contains a learning model **408** which is a mathematical artifact created by a machine learning training system **410** that generates predictions by finding patterns in one or more training data sets. In some embodiments, the machine learning subsystem **406** resides in the transport node **402**. In other embodiments, the machine learning subsystem **406** resides outside of the transport node **402**.

[0087] The transport node **402** sends data from the one or more sensors **404** to the machine learning subsystem **406**. The machine learning subsystem **406** provides the one or more sensor **404** data to the learning model **408** which returns one or more predictions. The machine learning subsystem **406** sends one or more instructions to the transport node **402** based on the predictions from the learning model **408**.

[0088] In a further embodiment, the transport node **402** may send the one or more sensor **404** data to the machine learning training system **410**. In yet another embodiment, the machine learning subsystem **406** may sent the sensor **404** data to the machine learning subsystem **410**. One or more of the applications, features, steps, solutions, etc., described and/or depicted herein may utilize the machine learning network **400** as described herein.

[0089] FIG. 5A illustrates an example vehicle configuration **500** for managing database transactions associated with a vehicle, according to example embodiments. Referring to FIG. 5A, as a particular transport/vehicle **525** is engaged in transactions (e.g., vehicle service, dealer transactions, delivery/pickup, transportation services, etc.), the vehicle may receive assets **510** and/or expel/transfer assets **512** according to a transaction(s). A transport processor **526** resides in the vehicle **525** and communication exists between the transport processor **526**, a database **530**, a transport processor **526** and the transaction module **520**. The transaction module **520** may record information, such as assets, parties, credits, service descriptions, date, time, location, results, notifications, unexpected events, etc. Those transactions in the transaction module **520** may be replicated into a database **530**. The database **530** can be one of a SQL database, an RDBMS, a relational database, a non-relational database, a blockchain, a distributed ledger, and may be on board the transport, may be off board the transport, may be accessible directly and/or through a network, or be accessible to the transport.

[0090] FIG. 5B illustrates an example vehicle configuration **550** for managing database transactions conducted among various vehicles, according to example embodiments. The vehicle **525** may engage with another vehicle **508** to perform various actions such as to share, transfer, acquire service calls, etc. when the vehicle has reached a status where the services need to be shared with another vehicle. For example, the vehicle **508** may be due for a battery charge and/or may have an issue with a tire and may be in route to pick up a package for delivery. A transport processor **528** resides in the vehicle **508** and communication exists between the transport processor **528**, a database **554**, a transport processor **528** and the transaction module **552**. The vehicle **508** may notify another vehicle **525** which is in its network and which operates on its blockchain member service. A transport processor **526** resides in the vehicle **525** and communication exists between the transport processor **526**, a database **530**, the transport processor **526** and a transaction

module **520**. The vehicle **525** may then receive the information via a wireless communication request to perform the package pickup from the vehicle **508** and/or from a server (not shown). The transactions are logged in the transaction modules **552** and **520** of both vehicles. The credits are transferred from vehicle **508** to vehicle **525** and the record of the transferred service is logged in the database **530/554** assuming that the blockchains are different from one another, or, are logged in the same blockchain used by all members. The database **554** can be one of a SQL database, an RDBMS, a relational database, a non-relational database, a blockchain, a distributed ledger, and may be on board the transport, may be off board the transport, may be accessible directly and/or through a network.

[0091] FIG. **6A** illustrates a blockchain architecture configuration **600**, according to example embodiments. Referring to FIG. **6A**, the blockchain architecture **600** may include certain blockchain elements, for example, a group of blockchain member nodes **602-606** as part of a blockchain group **610**. In one example embodiment, a permissioned blockchain is not accessible to all parties but only to those members with permissioned access to the blockchain data. The blockchain nodes participate in a number of activities, such as blockchain entry addition and validation process (consensus). One or more of the blockchain nodes may endorse entries based on an endorsement policy and may provide an ordering service for all blockchain nodes. A blockchain node may initiate a blockchain action (such as an authentication) and seek to write to a blockchain immutable ledger stored in the blockchain, a copy of which may also be stored on the underpinning physical infrastructure.

[0092] The blockchain transactions **620** are stored in memory of computers as the transactions are received and approved by the consensus model dictated by the members' nodes. Approved transactions **626** are stored in current blocks of the blockchain and committed to the blockchain via a committal procedure which includes performing a hash of the data contents of the transactions in a current block and referencing a previous hash of a previous block. Within the blockchain, one or more smart contracts **630** may exist that define the terms of transaction agreements and actions included in smart contract executable application code **632**, such as registered recipients, vehicle features, requirements, permissions, sensor thresholds, etc. The code may be configured to identify whether requesting entities are registered to receive vehicle services, what service features they are entitled/required to receive given their profile statuses and whether to monitor their actions in subsequent events. For example, when a service event occurs and a user is riding in the vehicle, the sensor data monitoring may be triggered, and a certain parameter, such as a vehicle charge level, may be identified as being above/below a particular threshold for a particular period of time, then the result may be a change to a current status which requires an alert to be sent to the managing party (i.e., vehicle owner, vehicle operator, server, etc.) so the service can be identified and stored for reference. The vehicle sensor data collected may be based on types of sensor data used to collect information about vehicle's status. The sensor data may also be the basis for the vehicle event data **634**, such as a location(s) to be traveled, an average speed, a top speed, acceleration rates, whether there were any collisions, was the expected route taken, what is the next endpoint, whether safety measures are in place, whether the vehicle has enough charge/fuel, etc. All such information may be the basis of smart contract terms **630**, which are then stored in a blockchain. For example, sensor thresholds stored in the smart contract can be used as the basis for whether a detected service is necessary and when and where the service should be performed.

[0093] FIG. **6B** illustrates a shared ledger configuration, according to example embodiments. Referring to FIG. **6B**, the blockchain logic example **640** includes a blockchain application interface **642** as an application programming interface or plug-in application that links to the computing device and execution platform for a particular transaction. The blockchain configuration **640** may include one or more applications which are linked to application programming interfaces (APIs) to access and execute stored program/application code (e.g., smart contract executable code, smart contracts, etc.) which can be created according to a customized configuration sought by

participants and can maintain their own state, control their own assets, and receive external information. This can be deployed as an entry and installed, via appending to the distributed ledger, on all blockchain nodes.

[0094] The smart contract application code **644** provides a basis for the blockchain transactions by establishing application code which when executed causes the transaction terms and conditions to become active. The smart contract **630**, when executed, causes certain approved transactions **626** to be generated, which are then forwarded to the blockchain platform **652**. The platform includes a security/authorization **658**, computing devices which execute the transaction management **656** and a storage portion **654** as a memory that stores transactions and smart contracts in the blockchain.

[0095] The blockchain platform may include various layers of blockchain data, services (e.g., cryptographic trust services, virtual execution environment, etc.), and underpinning physical computer infrastructure that may be used to receive and store new entries and provide access to auditors which are seeking to access data entries. The blockchain may expose an interface that provides access to the virtual execution environment necessary to process the program code and engage the physical infrastructure. Cryptographic trust services may be used to verify entries such as asset exchange entries and keep information private.

[0096] The blockchain architecture configuration of FIGS. **6A** and **6B** may process and execute program/application code via one or more interfaces exposed, and services provided, by the blockchain platform. As a non-limiting example, smart contracts may be created to execute reminders, updates, and/or other notifications subject to the changes, updates, etc. The smart contracts can themselves be used to identify rules associated with authorization and access requirements and usage of the ledger. For example, the information may include a new entry, which may be processed by one or more processing entities (e.g., processors, virtual machines, etc.) included in the blockchain layer. The result may include a decision to reject or approve the new entry based on the criteria defined in the smart contract and/or a consensus of the peers. The physical infrastructure may be utilized to retrieve any of the data or information described herein.

[0097] Within smart contract executable code, a smart contract may be created via a high-level application and programming language, and then written to a block in the blockchain. The smart contract may include executable code which is registered, stored, and/or replicated with a blockchain (e.g., distributed network of blockchain peers). An entry is an execution of the smart contract code which can be performed in response to conditions associated with the smart contract being satisfied. The executing of the smart contract may trigger a trusted modification(s) to a state of a digital blockchain ledger. The modification(s) to the blockchain ledger caused by the smart contract execution may be automatically replicated throughout the distributed network of blockchain peers through one or more consensus protocols.

[0098] The smart contract may write data to the blockchain in the format of key-value pairs. Furthermore, the smart contract code can read the values stored in a blockchain and use them in application operations. The smart contract code can write the output of various logic operations into the blockchain. The code may be used to create a temporary data structure in a virtual machine or other computing platform. Data written to the blockchain can be public and/or can be encrypted and maintained as private. The temporary data that is used/generated by the smart contract is held in memory by the supplied execution environment, then deleted once the data needed for the blockchain is identified.

[0099] A smart contract executable code may include the code interpretation of a smart contract, with additional features. As described herein, the smart contract executable code may be program code deployed on a computing network, where it is executed and validated by chain validators together during a consensus process. The smart contract executable code receives a hash and retrieves from the blockchain a hash associated with the data template created by use of a previously stored feature extractor. If the hashes of the hash identifier and the hash created from the stored identifier template data match, then the smart contract executable code sends an

authorization key to the requested service. The smart contract executable code may write to the blockchain data associated with the cryptographic details.

[0100] FIG. 6C illustrates a blockchain configuration for storing blockchain transaction data, according to example embodiments. Referring to FIG. 6C, the example configuration **660** provides for the vehicle **662**, the user device **664** and a server **666** sharing information with a distributed ledger (i.e., blockchain) **668**. The server may represent a service provider entity inquiring with a vehicle service provider to share user profile rating information in the event that a known and established user profile is attempting to rent a vehicle with an established rated profile. The server **666** may be receiving and processing data related to a vehicle's service requirements. As the service events occur, such as the vehicle sensor data indicates a need for fuel/charge, a maintenance service, etc., a smart contract may be used to invoke rules, thresholds, sensor information gathering, etc., which may be used to invoke the vehicle service event. The blockchain transaction data **670** is saved for each transaction, such as the access event, the subsequent updates to a vehicle's service status, event updates, etc. The transactions may include the parties, the requirements (e.g., 18 years of age, service eligible candidate, valid driver's license, etc.), compensation levels, the distance traveled during the event, the registered recipients permitted to access the event and host a vehicle service, rights/permissions, sensor data retrieved during the vehicle event operation to log details of the next service event and identify a vehicle's condition status, and thresholds used to make determinations about whether the service event was completed and whether the vehicle's condition status has changed.

[0101] FIG. 6D illustrates blockchain blocks **680** that can be added to a distributed ledger, according to example embodiments, and contents of block structures **682A** to **682n**. Referring to FIG. 6D, clients (not shown) may submit entries to blockchain nodes to enact activity on the blockchain. As an example, clients may be applications that act on behalf of a requester, such as a device, person or entity to propose entries for the blockchain. The plurality of blockchain peers (e.g., blockchain nodes) may maintain a state of the blockchain network and a copy of the distributed ledger. Different types of blockchain nodes/peers may be present in the blockchain network including endorsing peers which simulate and endorse entries proposed by clients and committing peers which verify endorsements, validate entries, and commit entries to the distributed ledger. In this example, the blockchain nodes may perform the role of endorser node, committer node, or both.

[0102] The instant system includes a blockchain which stores immutable, sequenced records in blocks, and a state database (current world state) maintaining a current state of the blockchain. One distributed ledger may exist per channel and each peer maintains its own copy of the distributed ledger for each channel of which they are a member. The instant blockchain is an entry log, structured as hash-linked blocks where each block contains a sequence of N entries. Blocks may include various components such as those shown in FIG. 6D. The linking of the blocks may be generated by adding a hash of a prior block's header within a block header of a current block. In this way, all entries on the blockchain are sequenced and cryptographically linked together preventing tampering with blockchain data without breaking the hash links. Furthermore, because of the links, the latest block in the blockchain represents every entry that has come before it. The instant blockchain may be stored on a peer file system (local or attached storage), which supports an append-only blockchain workload.

[0103] The current state of the blockchain and the distributed ledger may be stored in the state database. Here, the current state data represents the latest values for all keys ever included in the chain entry log of the blockchain. Smart contract executable code invocations execute entries against the current state in the state database. To make these smart contract executable code interactions extremely efficient, the latest values of all keys are stored in the state database. The state database may include an indexed view into the entry log of the blockchain, it can therefore be regenerated from the chain at any time. The state database may automatically get recovered (or

generated if needed) upon peer startup, before entries are accepted.

[0104] Endorsing nodes receive entries from clients and endorse the entry based on simulated results. Endorsing nodes hold smart contracts which simulate the entry proposals. When an endorsing node endorses an entry, the endorsing nodes creates an entry endorsement which is a signed response from the endorsing node to the client application indicating the endorsement of the simulated entry. The method of endorsing an entry depends on an endorsement policy which may be specified within smart contract executable code. An example of an endorsement policy is “the majority of endorsing peers must endorse the entry.” Different channels may have different endorsement policies. Endorsed entries are forward by the client application to an ordering service. [0105] The ordering service accepts endorsed entries, orders them into a block, and delivers the blocks to the committing peers. For example, the ordering service may initiate a new block when a threshold of entries has been reached, a timer times out, or another condition. In this example, blockchain node is a committing peer that has received a data block **682A** for storage on the blockchain. The ordering service may be made up of a cluster of orderers. The ordering service does not process entries, smart contracts, or maintain the shared ledger. Rather, the ordering service may accept the endorsed entries and specifies the order in which those entries are committed to the distributed ledger. The architecture of the blockchain network may be designed such that the specific implementation of ‘ordering’ (e.g., Solo, Kafka, BFT, etc.) becomes a pluggable component.

[0106] Entries are written to the distributed ledger in a consistent order. The order of entries is established to ensure that the updates to the state database are valid when they are committed to the network. Unlike a cryptocurrency blockchain system (e.g., Bitcoin, etc.) where ordering occurs through the solving of a cryptographic puzzle, or mining, in this example the parties of the distributed ledger may choose the ordering mechanism that best suits that network.

[0107] Referring to FIG. 6D, a block **682A** (also referred to as a data block) that is stored on the blockchain and/or the distributed ledger may include multiple data segments such as a block header **684A** to **684n**, transaction specific data **686A** to **686n**, and block metadata **688A** to **688n**. It should be appreciated that the various depicted blocks and their contents, such as block **682A** and its contents are merely for purposes of an example and are not meant to limit the scope of the example embodiments. In some cases, both the block header **684A** and the block metadata **688A** may be smaller than the transaction specific data **686A** which stores entry data; however, this is not a requirement. The block **682A** may store transactional information of N entries (e.g., 100, 500, 1000, 2000, 3000, etc.) within the block data **690A** to **690n**. The block **682A** may also include a link to a previous block (e.g., on the blockchain) within the block header **684A**. In particular, the block header **684A** may include a hash of a previous block's header. The block header **684A** may also include a unique block number, a hash of the block data **690A** of the current block **682A**, and the like. The block number of the block **682A** may be unique and assigned in an incremental/sequential order starting from zero. The first block in the blockchain may be referred to as a genesis block which includes information about the blockchain, its members, the data stored therein, etc.

[0108] The block data **690A** may store entry information of each entry that is recorded within the block. For example, the entry data may include one or more of a type of the entry, a version, a timestamp, a channel ID of the distributed ledger, an entry ID, an epoch, a payload visibility, a smart contract executable code path (deploy tx), a smart contract executable code name, a smart contract executable code version, input (smart contract executable code and functions), a client (creator) identify such as a public key and certificate, a signature of the client, identities of endorsers, endorser signatures, a proposal hash, smart contract executable code events, response status, namespace, a read set (list of key and version read by the entry, etc.), a write set (list of key and value, etc.), a start key, an end key, a list of keys, a Merkel tree query summary, and the like. The entry data may be stored for each of the N entries.

[0109] In some embodiments, the block data **690A** may also store transaction specific data **686A** which adds additional information to the hash-linked chain of blocks in the blockchain. Accordingly, the data **686A** can be stored in an immutable log of blocks on the distributed ledger. Some of the benefits of storing such data **686A** are reflected in the various embodiments disclosed and depicted herein. The block metadata **688A** may store multiple fields of metadata (e.g., as a byte array, etc.). Metadata fields may include signature on block creation, a reference to a last configuration block, an entry filter identifying valid and invalid entries within the block, last offset persisted of an ordering service that ordered the block, and the like. The signature, the last configuration block, and the orderer metadata may be added by the ordering service. Meanwhile, a committer of the block (such as a blockchain node) may add validity/invalidity information based on an endorsement policy, verification of read/write sets, and the like. The entry filter may include a byte array of a size equal to the number of entries in the block data **610A** and a validation code identifying whether an entry was valid/invalid.

[0110] The other blocks **682B** to **682n** in the blockchain also have headers, files, and values. However, unlike the first block **682A**, each of the headers **684A** to **684n** in the other blocks includes the hash value of an immediately preceding block. The hash value of the immediately preceding block may be just the hash of the header of the previous block or may be the hash value of the entire previous block. By including the hash value of a preceding block in each of the remaining blocks, a trace can be performed from the Nth block back to the genesis block (and the associated original file) on a block-by-block basis, as indicated by arrows **692**, to establish an auditable and immutable chain-of-custody.

[0111] The above embodiments may be implemented in hardware, in a computer program executed by a processor, in firmware, or in a combination of the above. A computer program may be embodied on a computer readable medium, such as a storage medium. For example, a computer program may reside in random access memory (“RAM”), flash memory, read-only memory (“ROM”), erasable programmable read-only memory (“EPROM”), electrically erasable programmable read-only memory (“EEPROM”), registers, hard disk, a removable disk, a compact disk read-only memory (“CD-ROM”), or any other form of storage medium known in the art.

[0112] An exemplary storage medium may be coupled to the processor such that the processor may read information from, and write information to, the storage medium. In the alternative, the storage medium may be integral to the processor. The processor and the storage medium may reside in an application specific integrated circuit (“ASIC”). In the alternative, the processor and the storage medium may reside as discrete components. For example, FIG. 7 illustrates an example computer system architecture **700**, which may represent or be integrated in any of the above-described components, etc.

[0113] FIG. 7 is not intended to suggest any limitation as to the scope of use or functionality of embodiments of the application described herein. Regardless, the computing node **700** is capable of being implemented and/or performing any of the functionality set forth hereinabove.

[0114] In computing node **700** there is a computer system/server **702**, which is operational with numerous other general purpose or special purpose computing system environments or configurations. Examples of well-known computing systems, environments, and/or configurations that may be suitable for use with computer system/server **702** include, but are not limited to, personal computer systems, server computer systems, thin clients, thick clients, hand-held or laptop devices, multiprocessor systems, microprocessor-based systems, set top boxes, programmable consumer electronics, network PCs, minicomputer systems, mainframe computer systems, and distributed cloud computing environments that include any of the above systems or devices, and the like.

[0115] Computer system/server **702** may be described in the general context of computer system-executable instructions, such as program modules, being executed by a computer system. Generally, program modules may include routines, programs, objects, components, logic, data

structures, and so on that perform particular tasks or implement particular abstract data types. Computer system/server **702** may be practiced in distributed cloud computing environments where tasks are performed by remote processing devices that are linked through a communications network. In a distributed cloud computing environment, program modules may be located in both local and remote computer system storage media including memory storage devices.

[0116] As shown in FIG. 7, computer system/server **702** in cloud computing node **700** is shown in the form of a general-purpose computing device. The components of computer system/server **702** may include, but are not limited to, one or more processors or processing units **704**, a system memory **706**, and a bus that couples various system components including system memory **706** to processor **704**.

[0117] The bus represents one or more of any of several types of bus structures, including a memory bus or memory controller, a peripheral bus, an accelerated graphics port, and a processor or local bus using any of a variety of bus architectures. By way of example, and not limitation, such architectures include Industry Standard Architecture (ISA) bus, Micro Channel Architecture (MCA) bus, Enhanced ISA (EISA) bus, Video Electronics Standards Association (VESA) local bus, and Peripheral Component Interconnects (PCI) bus.

[0118] Computer system/server **702** typically includes a variety of computer system readable media. Such media may be any available media that is accessible by computer system/server **702**, and it includes both volatile and non-volatile media, removable and non-removable media. System memory **706**, in one embodiment, implements the flow diagrams of the other figures. The system memory **706** can include computer system readable media in the form of volatile memory, such as random-access memory (RAM) **708** and/or cache memory **710**. Computer system/server **702** may further include other removable/non-removable, volatile/non-volatile computer system storage media. By way of example only, memory **706** can be provided for reading from and writing to a non-removable, non-volatile magnetic media (not shown and typically called a “hard drive”). Although not shown, a magnetic disk drive for reading from and writing to a removable, non-volatile magnetic disk (e.g., a “floppy disk”), and an optical disk drive for reading from or writing to a removable, non-volatile optical disk such as a CD-ROM, DVD-ROM or other optical media can be provided. In such instances, each can be connected to the bus by one or more data media interfaces. As will be further depicted and described below, memory **706** may include at least one program product having a set (e.g., at least one) of program modules that are configured to carry out the functions of various embodiments of the application.

[0119] Program/utility, having a set (at least one) of program modules, may be stored in memory **706** by way of example, and not limitation, as well as an operating system, one or more application programs, other program modules, and program data. Each of the operating system, one or more application programs, other program modules, and program data or some combination thereof, may include an implementation of a networking environment. Program modules generally carry out the functions and/or methodologies of various embodiments of the application as described herein.

[0120] As will be appreciated by one skilled in the art, aspects of the present application may be embodied as a system, method, or computer program product. Accordingly, aspects of the present application may take the form of an entirely hardware embodiment, an entirely software embodiment (including firmware, resident software, micro-code, etc.) or an embodiment combining software and hardware aspects that may all generally be referred to herein as a “circuit,” “module” or “system.” Furthermore, aspects of the present application may take the form of a computer program product embodied in one or more computer readable medium(s) having computer readable program code embodied thereon.

[0121] Computer system/server **702** may also communicate with one or more external devices via an I/O device **712** (such as an I/O adapter), which may include a keyboard, a pointing device, a display, a voice recognition module, etc., one or more devices that enable a user to interact with computer system/server **702**, and/or any devices (e.g., network card, modem, etc.) that enable

computer system/server **702** to communicate with one or more other computing devices. Such communication can occur via I/O interfaces of the device **712**. Still yet, computer system/server **702** can communicate with one or more networks such as a local area network (LAN), a general wide area network (WAN), and/or a public network (e.g., the Internet) via a network adapter. As depicted, device **712** communicates with the other components of computer system/server **702** via a bus. It should be understood that although not shown, other hardware and/or software components could be used in conjunction with computer system/server **702**. Examples, include, but are not limited to: microcode, device drivers, redundant processing units, external disk drive arrays, RAID systems, tape drives, and data archival storage systems, etc.

[0122] Although an exemplary embodiment of at least one of a system, method, and non-transitory computer readable medium has been illustrated in the accompanied drawings and described in the foregoing detailed description, it will be understood that the application is not limited to the embodiments disclosed, but is capable of numerous rearrangements, modifications, and substitutions as set forth and defined by the following claims. For example, the capabilities of the system of the various figures can be performed by one or more of the modules or components described herein or in a distributed architecture and may include a transmitter, receiver or pair of both. For example, all or part of the functionality performed by the individual modules, may be performed by one or more of these modules. Further, the functionality described herein may be performed at various times and in relation to various events, internal or external to the modules or components. Also, the information sent between various modules can be sent between the modules via at least one of: a data network, the Internet, a voice network, an Internet Protocol network, a wireless device, a wired device and/or via plurality of protocols. Also, the messages sent or received by any of the modules may be sent or received directly and/or via one or more of the other modules.

[0123] One skilled in the art will appreciate that a “system” could be embodied as a personal computer, a server, a console, a personal digital assistant (PDA), a cell phone, a tablet computing device, a smartphone or any other suitable computing device, or combination of devices. Presenting the above-described functions as being performed by a “system” is not intended to limit the scope of the present application in any way but is intended to provide one example of many embodiments. Indeed, methods, systems and apparatuses disclosed herein may be implemented in localized and distributed forms consistent with computing technology.

[0124] It should be noted that some of the system features described in this specification have been presented as modules, in order to more particularly emphasize their implementation independence. For example, a module may be implemented as a hardware circuit comprising custom very large-scale integration (VLSI) circuits or gate arrays, off-the-shelf semiconductors such as logic chips, transistors, or other discrete components. A module may also be implemented in programmable hardware devices such as field programmable gate arrays, programmable array logic, programmable logic devices, graphics processing units, or the like.

[0125] A module may also be at least partially implemented in software for execution by various types of processors. An identified unit of executable code may, for instance, comprise one or more physical or logical blocks of computer instructions that may, for instance, be organized as an object, procedure, or function. Nevertheless, the executables of an identified module need not be physically located together but may comprise disparate instructions stored in different locations which, when joined logically together, comprise the module and achieve the stated purpose for the module. Further, modules may be stored on a computer-readable medium, which may be, for instance, a hard disk drive, flash device, random access memory (RAM), tape, or any other such medium used to store data.

[0126] Indeed, a module of executable code could be a single instruction, or many instructions, and may even be distributed over several different code segments, among different programs, and across several memory devices. Similarly, operational data may be identified and illustrated herein

within modules and may be embodied in any suitable form and organized within any suitable type of data structure. The operational data may be collected as a single data set or may be distributed over different locations including over different storage devices, and may exist, at least partially, merely as electronic signals on a system or network.

[0127] It will be readily understood that the components of the application, as generally described and illustrated in the figures herein, may be arranged and designed in a wide variety of different configurations. Thus, the detailed description of the embodiments is not intended to limit the scope of the application as claimed but is merely representative of selected embodiments of the application.

[0128] One having ordinary skill in the art will readily understand that the above may be practiced with steps in a different order, and/or with hardware elements in configurations that are different than those which are disclosed. Therefore, although the application has been described based upon these preferred embodiments, it would be apparent to those of skill in the art that certain modifications, variations, and alternative constructions would be apparent.

[0129] While preferred embodiments of the present application have been described, it is to be understood that the embodiments described are illustrative only and the scope of the application is to be defined solely by the appended claims when considered with a full range of equivalents and modifications (e.g., protocols, hardware devices, software platforms etc.) thereto.

Claims

1. A method, comprising: transmitting, by a transport, status information related to the transport; based on the status information, transmitting, by the transport, an amount of energy that is available in a battery on the transport; based on the amount of energy, receiving, by the transport, a request to maneuver to a charging station; based on the request, maneuvering, by the transport, to the charging station; and providing, by the transport, the amount of energy to the charging station.
2. The method of claim 1, further comprising: identifying an environment at a location of the charging station based on an external temperature of the transport and a geographic position of the location.
3. The method of claim 1, further comprising: identifying an environment at a location of the charging station based on a current seasonal shading of the location and a current weather condition.
4. The method of claim 1, further comprising: scheduling the receiving of the amount of energy based on an energy cost associated with a location of the charging station.
5. The method of claim 1, further comprising: determining the amount of energy needed based on one or more of an external temperature at the transport, an estimated temperature at a location of the charging station, or an energy cost associated with the location.
6. The method of claim 1, further comprising: sending a state of a charge of the charging station from the charging station to the transport.
7. The method of claim 1, further comprising: executing a smart contract related to the amount of energy needed between the transport and the charging station.
8. A transport, comprising: a processor that, when executing instructions stored in a memory, is configured to: transmit status information related to the transport; based on the status information, transmit an amount of energy that is available in a battery on the transport; based on the amount of energy, receive a request to maneuver to a charging station; based on the request, maneuver to the charging station; and provide the amount of energy to the charging station.
9. The transport of claim 8, wherein the processor is further configured to: identify an environment at a location of the charging station based on an external temperature of the transport and a geographic position of the location.
10. The transport of claim 8, wherein the processor is further configured to: identify an

environment at a location is based on a current seasonal shading of the location and a current weather condition.

11. The transport of claim 8, wherein the processor is further configured to: schedule the amount of energy being received based on an energy cost associated with a location.

12. The transport of claim 8, wherein the processor is further configured to: determine the amount of energy needed based on one or more of an external temperature at the transport, an estimated temperature at a location of the charging station, or an energy cost associated with the location.

13. The transport of claim 8, wherein the processor is further configured to: send a state of a charge of the charging station from the charging station to the transport.

14. The transport of claim 8, wherein the processor is further configured to: execute a smart contract related to the amount of energy needed between the transport and the charging station.

15. A non-transitory computer-readable medium comprising instructions that, when executed by a processor, cause the processor to perform: transmitting status information related to a transport; based on the status information, transmitting an amount of energy that is available in a battery on the transport; based on the amount of energy, receiving a request to maneuver to a charging station; based on the request, maneuvering to the charging station; and providing the amount of energy to the charging station.

16. The non-transitory computer-readable medium of claim 15, wherein the instructions further cause the processor to perform: identifying an environment at a location of the charging station based on an external temperature of the transport and a geographic position of the location.

17. The non-transitory computer-readable medium of claim 15, wherein the instructions further cause the processor to perform: identifying an environment at a location of the charging station based on a current seasonal shading of the location and a current weather condition.

18. The non-transitory computer-readable medium of claim 15, wherein the instructions further cause the processor to perform: scheduling the receiving of the amount of energy based on an energy cost associated with a location.

19. The non-transitory computer-readable medium of claim 15, wherein the instructions further cause the processor to perform: determining the amount of energy needed based on one or more of an external temperature at the transport, an estimated temperature at a location of the charging station, or an energy cost associated with the location.

20. The non-transitory computer-readable medium of claim 15, wherein the instructions further cause the processor to perform: sending a state of a charge of the charging station from the charging station to the transport.
